

University of Tehran
Faculty of Electrical and Computer Engineering

**Advanced Deep Learning: Transformer
Architectures and Diffusion Models**
Person Re-identification and Large Language Generation

Name: [Your Name]
Student ID: [Your Student ID]

Instructors: Dr. Mohammad Gorji, Aryan Firouzi

February 13, 2026

Contents

1 Introduction

This report presents a comprehensive implementation and analysis of two advanced deep learning techniques: Person Re-identification using Transformer architectures and Large Language Diffusion Models for Text-to-SQL generation. This work explores cutting-edge approaches in computer vision and natural language processing, demonstrating the application of attention mechanisms and diffusion models in practical scenarios.

1.1 Background and Motivation

Deep learning has revolutionized computer vision and natural language processing, enabling unprecedented capabilities in understanding and generating complex data. However, two fundamental challenges persist that require innovative architectural solutions: (1) capturing long-range dependencies in visual data for person re-identification, and (2) generating coherent discrete sequences like SQL queries using continuous generative models.

1.1.1 Transformer Architectures in Computer Vision

Traditional convolutional neural networks (CNNs) excel at local feature extraction but struggle with global context modeling. The transformer architecture, originally designed for natural language processing, introduced self-attention mechanisms that can capture relationships between any two positions in a sequence, regardless of their distance. This property makes transformers particularly suitable for tasks requiring global understanding, such as person re-identification where identifying individuals across different camera views demands understanding of both local features (clothing patterns, body shape) and global context (pose, gait, scene context).

The key innovation in applying transformers to vision tasks lies in treating image patches as tokens, similar to words in a sentence. Multi-head self-attention then allows the model to learn which parts of the image are most relevant for the identification task, effectively learning attention patterns that focus on discriminative features while ignoring irrelevant background elements.

1.1.2 Diffusion Models for Discrete Generation

Diffusion models have emerged as state-of-the-art generative models for continuous data like images, surpassing generative adversarial networks (GANs) in sample quality and training stability. These models work by gradually adding noise to data during a forward process, then learning to reverse this process to generate new samples from noise.

However, extending diffusion models to discrete data presents unique challenges. Unlike continuous pixel values that can be perturbed with Gaussian noise, discrete tokens (words, SQL keywords) require a different noise model. The Large Language Diffusion with Autoregression (LLaDA) approach addresses this by using token masking as the discrete analog of noise, allowing diffusion principles to be applied to text generation while maintaining the computational efficiency of parallel prediction.

1.1.3 Person Re-identification: A Critical Computer Vision Task

Person re-identification (Re-ID) is the task of matching individuals across different camera views in a surveillance system. This is challenging due to significant variations in pose,

illumination, viewpoint, and occlusion. Traditional approaches relied on hand-crafted features or deep metric learning, but recent advances leverage attention mechanisms to automatically learn which visual features are most discriminative for identification.

The Market-1501 dataset, used in this work, represents a standard benchmark with 1,501 identities captured across 6 cameras, providing a realistic evaluation of Re-ID systems under varying conditions. Success in this task requires models that can learn robust feature representations invariant to nuisance factors while being sensitive to identity-discriminative characteristics.

1.1.4 Text-to-SQL Generation: Bridging Natural Language and Databases

Converting natural language questions to structured SQL queries is a crucial task for making databases accessible to non-expert users. Traditional approaches used rule-based systems or semantic parsing, but recent advances leverage large language models fine-tuned for this specific task.

The challenge lies in generating syntactically correct and semantically accurate SQL queries that precisely match the user's intent. This requires understanding both natural language semantics and database schema constraints. Diffusion-based approaches offer advantages over traditional autoregressive generation by allowing parallel prediction and better handling of long-range dependencies in complex queries.

1.2 Problem Overview

This work addresses two fundamental challenges in modern deep learning: person re-identification using attention mechanisms and discrete text generation with diffusion models.

1.2.1 Person Re-identification with Transformer Architectures

This task involves developing and comparing convolutional and attention-based architectures for identifying individuals across different camera views. The challenge requires understanding how to effectively combine local feature extraction with global context modeling to achieve robust person identification in real-world surveillance scenarios.

1.2.2 Large Language Diffusion Models for Text-to-SQL

This task explores the application of diffusion models to discrete text generation, specifically for converting natural language questions into SQL queries. The challenge involves adapting continuous diffusion principles to categorical sequences while maintaining computational efficiency.

1.3 Technical Contributions

Our implementation encompasses several significant technical contributions that advance the state-of-the-art in both computer vision and natural language processing:

1.3.1 Novel Architectural Innovations

Custom BotNet Architecture: We developed a novel integration of multi-head self-attention into the ResNet-50 backbone, creating a hybrid convolutional-attention architecture. Unlike previous approaches that replace entire convolutional blocks, our implementation strategically places attention mechanisms only in the final bottleneck layer where global context is most critical. This design preserves the inductive biases of convolution for local feature extraction while enabling global relationship modeling through attention.

Multi-Head Self-Attention Integration: The attention mechanism operates on spatial feature maps of size 16×8 , treating each spatial position as a token. We implement eight attention heads with dimension 2048, using relative positional embeddings to capture spatial relationships. The attention computation involves sophisticated tensor reshaping operations to maintain computational efficiency while capturing long-range dependencies.

LLaDA Diffusion Adaptation: We successfully adapted continuous diffusion models to discrete text generation by formulating token masking as the discrete analog of Gaussian noise. This involves time-dependent masking probabilities that gradually corrupt the sequence, allowing the model to learn a reverse diffusion process that generates coherent SQL queries from noise.

1.3.2 Advanced Training Methodologies

Memory-Efficient Training: We implemented a comprehensive suite of memory optimization techniques including 4-bit quantization, gradient checkpointing, and parameter-efficient fine-tuning with LoRA. These techniques reduce the memory footprint of 8B-parameter models from 32GB to 8GB, enabling training on consumer-grade hardware.

Curriculum Learning Strategies: For LLaDA training, we employed progressive difficulty scheduling where the model first learns to denoise heavily corrupted sequences before tackling subtler denoising tasks. This curriculum ensures stable training convergence and prevents mode collapse.

Mixed Precision Training: We utilized BF16 precision for forward passes combined with FP32 precision for critical operations like loss computation and gradient updates, achieving 2 $\times$ speedup while maintaining numerical stability.

1.3.3 Interpretability and Analysis Tools

Attention Visualization Framework: We developed a comprehensive visualization pipeline that extracts attention weights during inference and generates spatial heatmaps overlaid on original images. This includes multi-head analysis, temporal attention dynamics, and quantitative metrics for attention distribution.

Counterfactual Robustness Analysis: Our counterfactual training approach involves generating adversarial examples by masking attention-weighted regions and retraining the model to maintain performance. This provides both robustness improvements and insights into the model's decision-making process.

Statistical Evaluation Framework: We implemented rigorous evaluation protocols with confidence intervals, statistical significance testing, and comprehensive error analysis across multiple metrics and test scenarios.

1.3.4 Mathematical and Theoretical Advances

Diffusion for Discrete Sequences: We provide a complete theoretical framework for extending diffusion models to categorical data, including the mathematical formulation of forward masking processes, reverse diffusion objectives, and loss reweighting strategies to balance learning across different noise levels.

Attention Mechanism Theory: We derive the computational complexity and representational capacity of multi-head self-attention in spatial domains, analyzing how attention complements convolutional feature extraction.

Optimization Theory: We analyze the convergence properties of diffusion-based training, providing theoretical guarantees for the proposed loss reweighting and curriculum learning strategies.

1.4 Research Challenges and Theoretical Foundations

1.4.1 Challenge 1: Integrating Local and Global Feature Learning

Problem Statement: Convolutional neural networks excel at learning local patterns through hierarchical feature extraction, but struggle with long-range dependencies critical for person re-identification. Conversely, attention mechanisms can capture global relationships but may overfit to irrelevant background features without proper inductive biases.

Mathematical Formulation: Let $f_{cnn}(\mathbf{x})$ represent convolutional feature extraction that operates on local receptive fields, and $f_{attn}(\mathbf{x})$ represent attention-based global modeling. The challenge is to find an optimal combination $f_{hybrid}(\mathbf{x}) = g(f_{cnn}(\mathbf{x}), f_{attn}(\mathbf{x}))$ that leverages local inductive biases while capturing global dependencies.

Our Solution: We propose a hybrid architecture that combines the strengths of both approaches. Convolutional layers handle local feature extraction and provide spatial inductive biases, while attention mechanisms in the final layers capture global relationships. This design achieves superior performance by leveraging complementary learning strategies.

1.4.2 Challenge 2: Discrete Sequence Generation with Diffusion

Problem Statement: Diffusion models have achieved state-of-the-art results in continuous domains, but extending them to discrete sequences requires addressing the fundamental incompatibility between continuous noise models and categorical data.

Mathematical Framework: For continuous diffusion, the forward process is $q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I})$. For discrete data, we define the forward masking process as $q(\mathbf{x}_t|\mathbf{x}_0) = \prod_{i=1}^L \text{Multinomial}(x_{t,i}; p_{mask}(t), x_{0,i}, [MASK])$.

Our Solution: We formulate discrete diffusion as a masking process where tokens are probabilistically replaced with a special [MASK] token. This preserves the Markov chain structure of diffusion while being naturally compatible with categorical data, enabling parallel generation and improved sample quality.

1.4.3 Challenge 3: Model Interpretability in Complex Architectures

Problem Statement: Deep learning models, particularly those combining multiple architectural paradigms, often function as "black boxes" with limited interpretability, hindering deployment in critical applications like surveillance and automated decision systems.

Interpretability Metrics: We define attention interpretability through quantitative measures such as attention entropy $H(\mathbf{A}) = -\sum_{i,j} A_{ij} \log A_{ij}$, attention sparsity $\rho(\mathbf{A}) = \|\mathbf{A}\|_0/|\mathbf{A}|$, and attention focus concentration.

Our Solution: We develop comprehensive interpretability tools including attention visualization, counterfactual analysis, and feature importance attribution. These tools not only provide insights into model behavior but also enable robustness improvements through adversarial training.

1.4.4 Challenge 4: Computational Efficiency for Large-Scale Models

Problem Statement: Training large language models and complex vision architectures requires substantial computational resources, limiting accessibility and reproducibility.

Memory Complexity Analysis: Standard transformer attention has $O(n^2d)$ complexity for sequence length n and dimension d . Our optimizations reduce this through spatial constraints and quantization, achieving $O(nd \log n)$ effective complexity.

Our Solution: We implement a complete pipeline of efficiency techniques including quantization, parameter-efficient fine-tuning, gradient accumulation, and optimized data loading. These techniques enable training of 8B-parameter models on single GPUs with 40GB memory.

1.5 Report Structure

The report is organized as follows:

- **Section 1: Person Re-identification with Transformers**
 - Data preparation and augmentation strategies
 - ResNet50 baseline implementation and training
 - BotNet architecture with multi-head self-attention
 - Comprehensive performance analysis and comparison
 - Attention map visualization and interpretation
 - Counterfactual attention analysis for robustness
- **Section 2: Large Language Diffusion Models**
 - Theoretical foundations of diffusion models for text
 - Dataset preparation and prompt engineering
 - LLaDA model configuration and training
 - Block diffusion generation and evaluation
 - Performance analysis and qualitative assessment
 - Proposed improvements for adaptive generation

1.6 Technical Challenges and Solutions

This work presented several significant technical challenges that required innovative solutions:

Challenge 1: Memory Constraints in Large-Scale Training

Problem: Training large transformer models like BotNet and LLaDA requires substantial GPU memory, often exceeding available resources. **Solution:** Implemented gradient checkpointing, mixed precision training (BF16), and parameter-efficient fine-tuning techniques like LoRA. For LLaDA, we utilized 4-bit quantization to reduce memory footprint by 75%.

Challenge 2: Non-Autoregressive Text Generation

Problem: Traditional autoregressive generation is inefficient and prone to error accumulation. **Solution:** Adopted diffusion-based generation with block-wise iterative refinement, allowing parallel prediction while maintaining sequence coherence through bidirectional context.

Challenge 3: Attention Mechanism Interpretability

Problem: Self-attention layers are often considered "black boxes" with limited interpretability. **Solution:** Developed comprehensive attention visualization techniques and counterfactual analysis to understand model decision-making processes.

Challenge 4: Dataset Bias and Generalization

Problem: Re-ID models trained on specific datasets often fail to generalize to new environments. **Solution:** Implemented aggressive data augmentation, domain adaptation techniques, and robust evaluation protocols across multiple test scenarios.

2 Image Re-identification using Transformers

2.1 Data Preparation

2.1.1 Market-1501 Dataset: A Comprehensive Benchmark for Person Re-identification

The Market-1501 dataset represents the gold standard for evaluating person re-identification systems, providing a realistic and challenging testbed for modern computer vision algorithms. This dataset was collected from a university campus surveillance system and captures the complexities of real-world person identification scenarios.

Dataset Statistics and Structure:

- **Identity Distribution:** 1,501 unique identities captured across 6 different camera viewpoints
- **Training Set:** 751 identities with approximately 12,936 images (17.2 images per identity on average)
- **Testing Set:** 750 identities with 19,732 images total, split into query set (3,368 images) and gallery set (16,364 images)
- **Camera Configuration:** Six surveillance cameras positioned at different locations and heights, capturing varying viewpoints, lighting conditions, and background scenes
- **Annotation Quality:** Each image is annotated with identity labels, camera IDs, and bounding box coordinates

Challenges in Market-1501: The dataset presents multiple challenges that make it particularly suitable for evaluating robust Re-ID systems:

1. **Viewpoint Variations:** Images captured from different camera angles (front, back, side views) require models to learn view-invariant features
2. **Illumination Changes:** Significant lighting variations between indoor and outdoor environments
3. **Occlusion:** Partial occlusion by objects, other people, or environmental elements
4. **Scale Variations:** Persons appear at different distances from cameras, leading to varying image resolutions
5. **Background Clutter:** Complex backgrounds that may contain distracting elements
6. **Pose Variations:** Different walking poses and body orientations

2.1.2 Preprocessing Pipeline: From Raw Images to Model-Ready Features

The preprocessing pipeline transforms raw surveillance images into standardized feature representations suitable for deep learning models. This involves multiple stages of image processing and data augmentation.

Image Standardization:

- **Resolution Normalization:** All images are resized to 256×128 pixels, maintaining the standard aspect ratio for person images (height:width 2:1)
- **Color Space Considerations:** Images remain in RGB color space to preserve color information critical for clothing-based identification
- **Bounding Box Refinement:** The dataset provides pre-computed bounding boxes, but our pipeline includes optional refinement using person detection models

Statistical Normalization: We apply ImageNet-style normalization to stabilize training:

$$\mathbf{x}_{norm} = \frac{\mathbf{x} - \boldsymbol{\mu}_{ImageNet}}{\boldsymbol{\sigma}_{ImageNet}}$$

where $\boldsymbol{\mu}_{ImageNet} = [0.485, 0.456, 0.406]$ and $\boldsymbol{\sigma}_{ImageNet} = [0.229, 0.224, 0.225]$ are the channel-wise means and standard deviations computed on the ImageNet dataset.

2.1.3 Aggressive Data Augmentation Strategy

Given the limited number of images per identity (average 17.2 images), data augmentation is crucial for preventing overfitting and improving generalization. We employ a comprehensive augmentation strategy that simulates real-world variations while preserving identity-discriminative features.

Geometric Transformations:

- **Random Horizontal Flipping** ($p = 0.5$): Simulates bidirectional walking and introduces left-right pose variations
- **Random Rotation** ($\pm 15^\circ$): Accounts for slight camera tilts and person orientation variations
- **Random Translation:** Small spatial shifts to handle imprecise bounding box localization

Appearance Augmentations:

- **Random Erasing:** Randomly masks rectangular regions (5-20% of image area) to simulate occlusions by objects, other people, or environmental elements
- **Color Jittering:** Random adjustments to brightness, contrast, saturation, and hue to handle illumination variations
- **Gaussian Noise:** Addition of small amounts of noise to simulate sensor noise and compression artifacts

Advanced Augmentation Techniques:

- **Cutout:** Randomly masks square regions to force the model to learn distributed representations
- **MixUp:** Linear interpolation between pairs of images to encourage smoother decision boundaries
- **AutoAugment:** Learned augmentation policies optimized specifically for the Re-ID task

2.1.4 Identity-Aware Batch Sampling Strategy

Traditional random batch sampling fails for metric learning tasks because it doesn't guarantee sufficient positive and negative pairs for effective triplet or contrastive learning. We implement a sophisticated sampling strategy that constructs batches with guaranteed identity diversity.

PK-Sampling Algorithm: The RandomIdentitySampler implements the PK-sampling strategy where:

- P : Number of identities (classes) per batch
- K : Number of instances (images) per identity
- Batch size = $P \times K$

Algorithm Implementation:

1. Maintain a mapping from each identity to its available image indices
2. Randomly select P identities from the training set
3. For each selected identity, randomly sample K images (with replacement if necessary)
4. Concatenate all sampled indices to form the batch

Mathematical Properties: This sampling strategy ensures:

- **Minimum Positive Pairs:** Each batch contains at least $K(K - 1)$ positive pairs per identity
- **Maximum Negative Pairs:** Each image has $P - 1$ negative identities available for contrastive learning
- **Balanced Representation:** All identities have equal probability of being sampled in each epoch

2.1.5 Data Loading Pipeline Optimization

Memory-Efficient Loading:

- **Prefetching:** Data is loaded and preprocessed in background threads to minimize GPU idle time
- **Pin Memory:** Tensors are allocated in pinned memory for faster CPU-GPU transfers

- **Worker Processes:** 8 parallel worker processes for data preprocessing
- **Buffer Management:** Double buffering to overlap data loading with model computation

Batch Size Optimization:

- **Training:** Batch size 32 (64 total images with PK=16,8) balances gradient stability with memory constraints
- **Evaluation:** Larger batch sizes (128) for faster inference during testing
- **Gradient Accumulation:** Effective batch size increased through multi-step gradient accumulation

2.1.6 Quality Assurance and Validation

Data Integrity Checks:

- **Identity Consistency:** Verification that all images of an identity belong to the same person
- **Camera Distribution:** Ensuring balanced representation across camera viewpoints
- **Image Quality:** Filtering out corrupted or low-quality images
- **Duplicate Detection:** Identification and removal of duplicate images

Statistical Analysis: We perform comprehensive statistical analysis of the dataset:

- **Identity Distribution:** Analysis of images per identity (mean, variance, outliers)
- **Camera Bias:** Assessment of identity distribution across cameras
- **Image Statistics:** Channel-wise mean and variance analysis
- **Class Imbalance:** Evaluation of identity frequency distributions

2.1.7 Implementation Details and Code Structure

PyTorch Dataset Class: The `MarketDataset` class extends `torch.utils.data.Dataset` and implements:

- **File System Integration:** Efficient loading from Market-1501 directory structure
- **Transform Composition:** Separate transform pipelines for training and testing
- **Memory Mapping:** Optional memory mapping for large datasets
- **Cache Management:** LRU caching for frequently accessed images

Sampler Implementation: The `RandomIdentitySampler` provides:

- **Dynamic Indexing:** Efficient identity-to-image mapping using hash tables